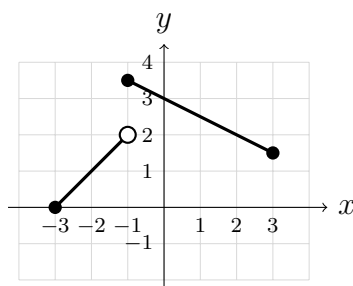


Math 133: Quiz 1 — Answer Key

Friday, September 11, 2026

1. Domain: $\{-2, 0, 1, 3\}$. Range: $\{2, 4, 5\}$. No: each input has exactly one output; different inputs may have the same output.
2. The domain of h is $(-\infty, 4) \cup (4, \infty)$. The domain of k is $[-2, \infty)$.
3. $C(10) = 50$: making 10 mugs costs a total of \$50. The \$30 is the fixed cost, incurred even when no mugs are made. The \$2 per mug is the additional cost for each mug made.
4. A line segment from $(-3, 0)$ (filled) to $(-1, 2)$ (open), and a line segment from $(-1, 3.5)$ (filled) to $(3, 1.5)$ (filled).



5.

x	-1	0	1	3
$F(x)$	-1	-2	3	1

Domain: $[-2, 4]$. Range: $[-2, 3]$. Increasing on $(0, 1)$; decreasing on $(-2, 0)$ and $(1, 4)$.